Mohideen AbdulKader ML
240701329

EXPERIMENT 6
Implementation of 10 Deluge Functions in Zoho Creator

1. Aim

To implement and execute ten basic Deluge functions in Zoho Creator and observe how scripting can be used for calculations, decision-making, repetition, and simple data-processing operations.

2. Introduction

Deluge (Data Enriched Language for the Universal Grid Environment) is the scripting language used across Zoho products for adding logic and automation. In Zoho Creator, Deluge statements can be used to perform calculations, evaluate conditions, repeat operations, and process values according to application requirements.

3. Objectives of the Study

- To become familiar with basic Deluge syntax and statements.
- To perform arithmetic operations using Deluge variables.
- To apply conditional statements for decision-making.
- To use iteration for repetitive calculations.
- To perform basic operations on numbers and strings and verify their outputs.

4. Procedure

1. Open Zoho Creator and enter the Deluge scripting environment.
2. Create or open an application in which the functions can be tested.
3. Create a separate function or script for each problem.
4. Declare the required variables and write the Deluge statements.
5. Use conditional statements or loops wherever the problem requires them.
6. Execute each script with suitable input values.
7. Observe the information displayed after execution.
8. Verify the result and save the completed functions.

5. Deluge Functions and Outputs

1. Sum of Two Numbers

Two numerical values are assigned to variables and combined using the addition operator. The calculated total is displayed through the execution message.

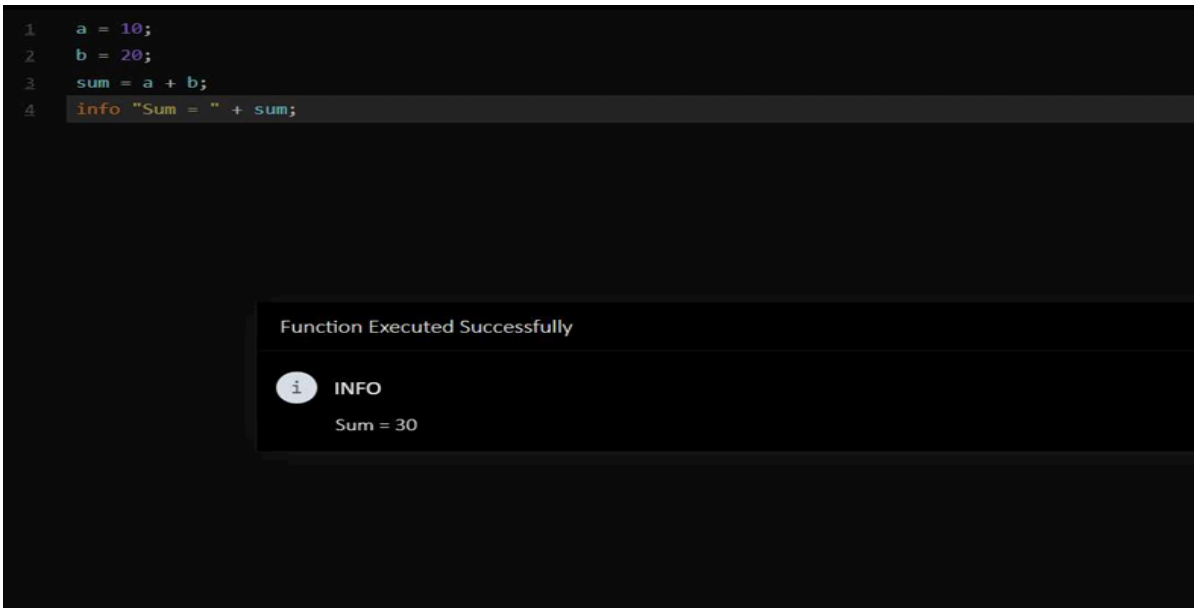


Figure 1: Deluge function for adding two numbers and its execution output.

2. Even or Odd Check

The input number is tested with the modulo operator. If the remainder after division by two is zero, the number is identified as even; otherwise it is identified as odd.

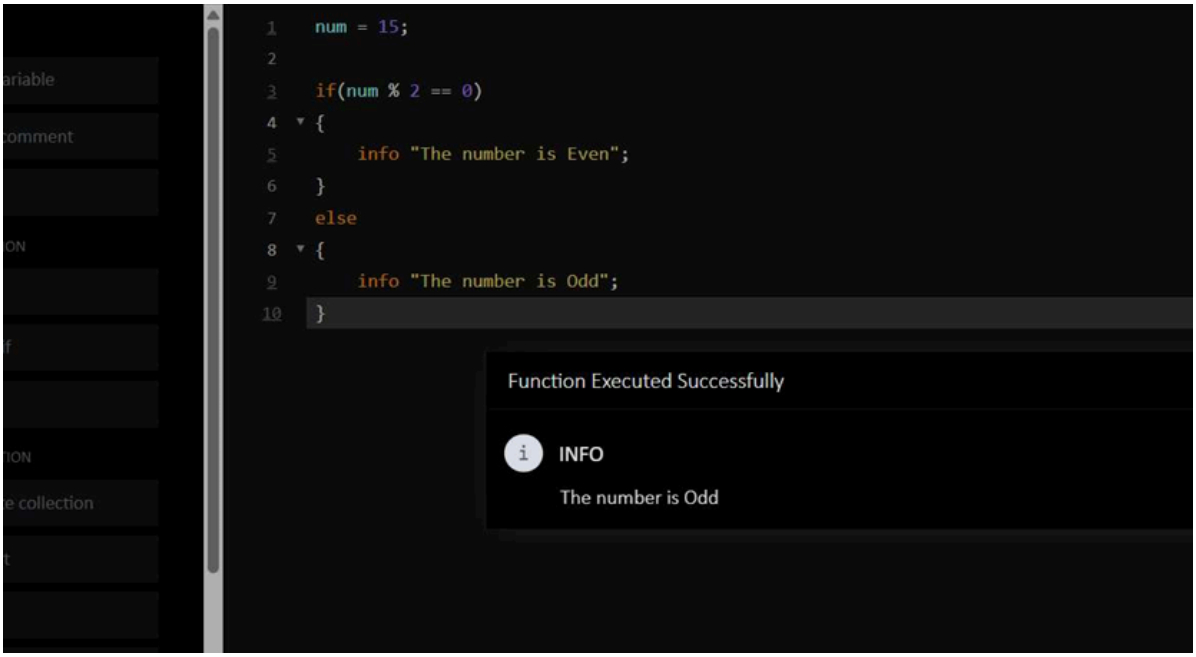


Figure 2: Deluge condition used to identify an even or odd number.

3. Largest Number

Conditional comparisons are used to examine the supplied values and determine which one has the greatest value.

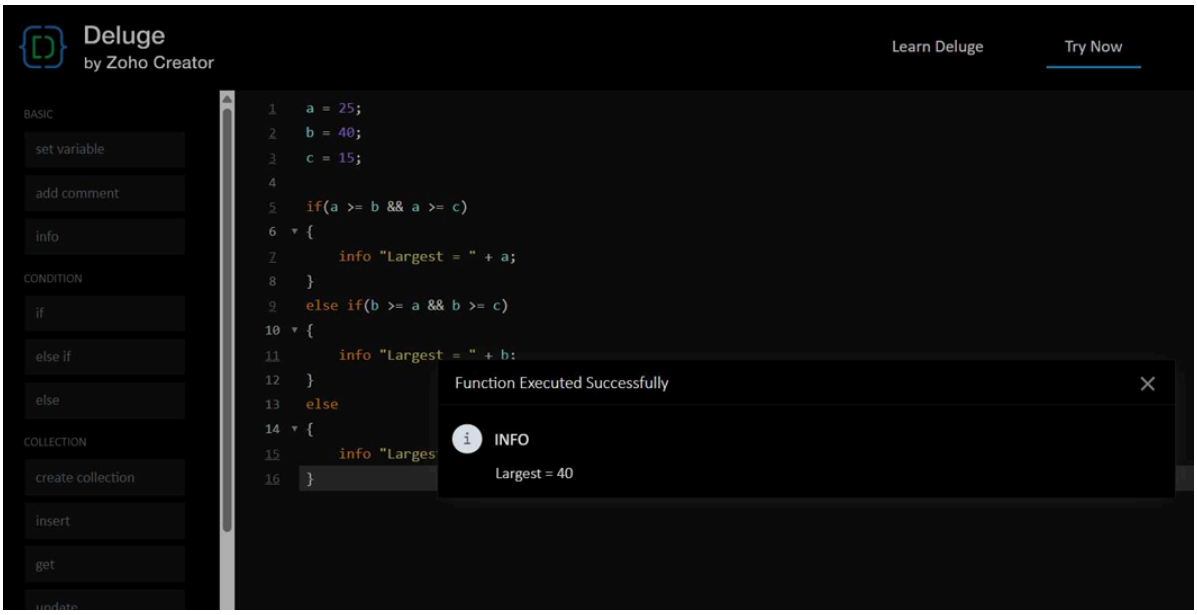


Figure 3: Deluge comparison logic for finding the largest value.

4. Positive, Negative or Zero

The value is compared with zero using conditional branches. The script reports whether the number is positive, negative, or exactly zero.



Figure 4: Deluge conditional logic for classifying a number.

5. Factorial

A running product is updated repeatedly using a loop to obtain the factorial of the selected number.

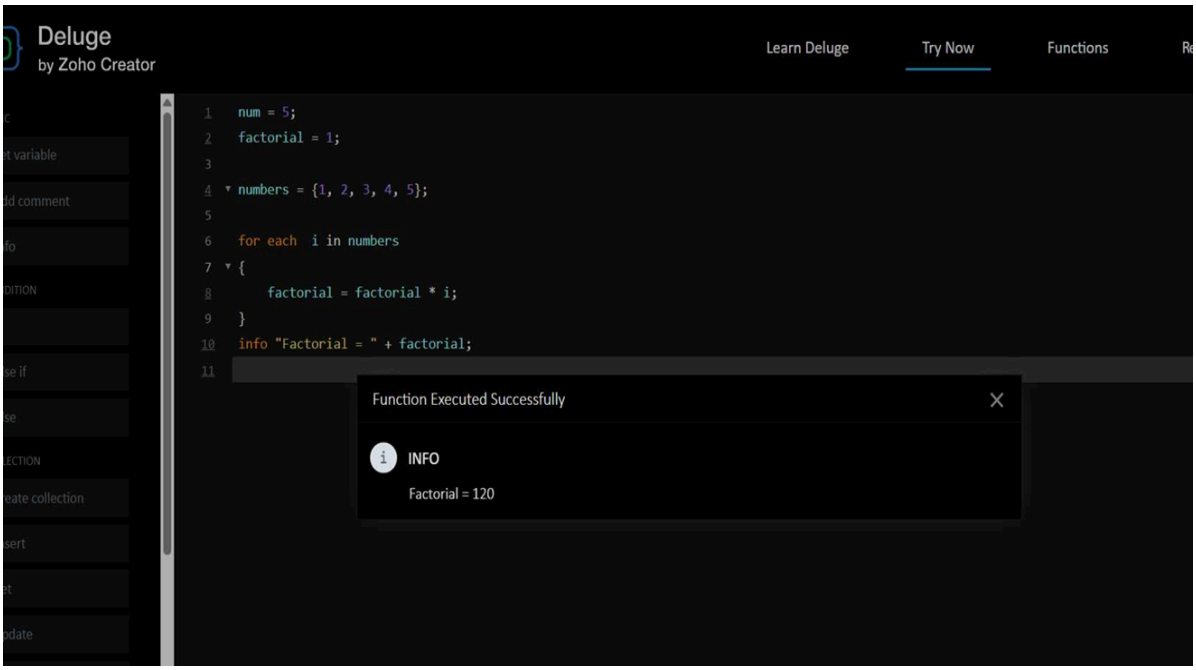


Figure 5: Deluge loop used for factorial calculation.

6. Multiplication Table

Iteration is used to multiply the selected number by successive values and display the resulting multiplication table.

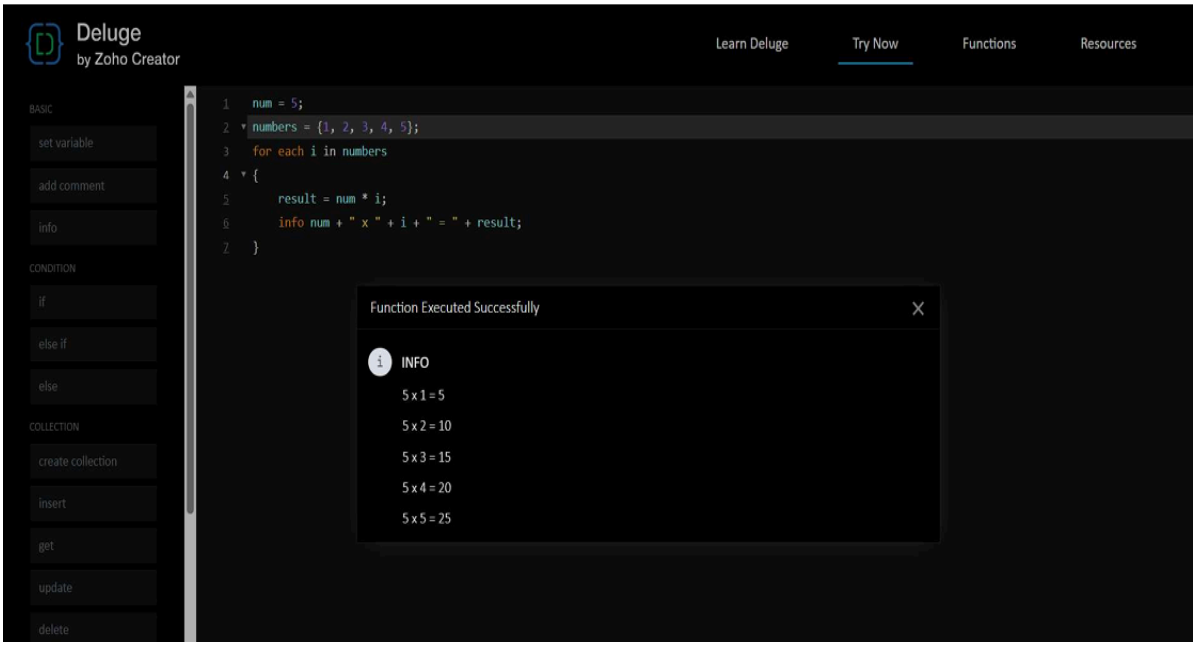


Figure 6: Deluge iteration for generating a multiplication table.

7. Prime Number Check

The script checks possible factors of the given number and uses the count of valid factors to determine whether the number is prime.

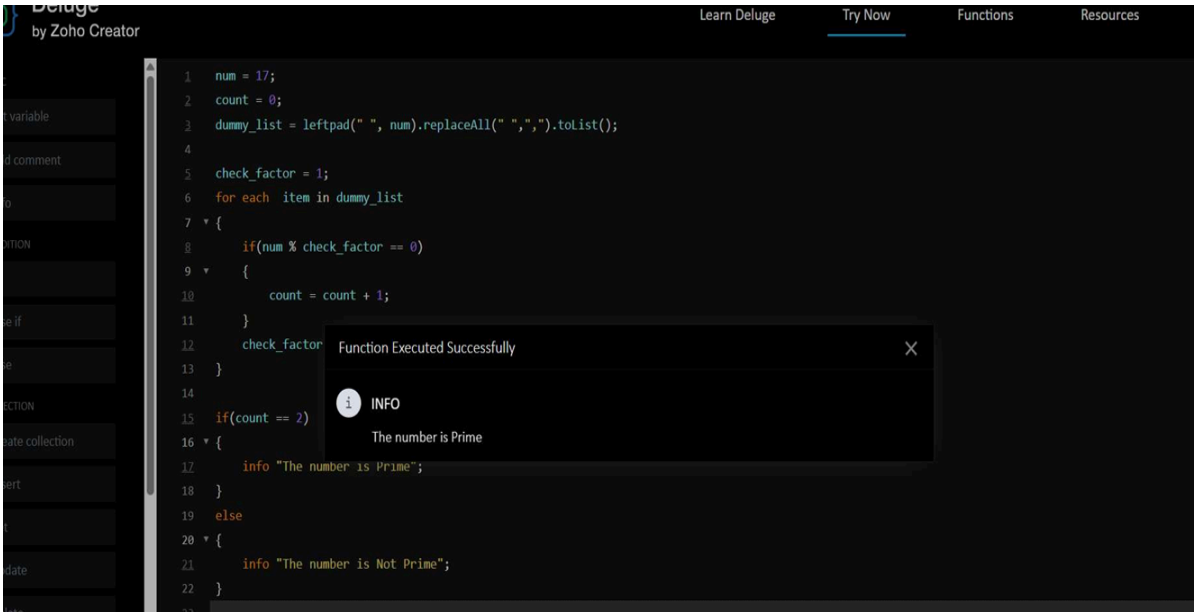


Figure 7: Deluge logic for checking whether a number is prime.

8. Reverse a Number

The digits of the supplied number are reversed and the resulting value is displayed as the output.

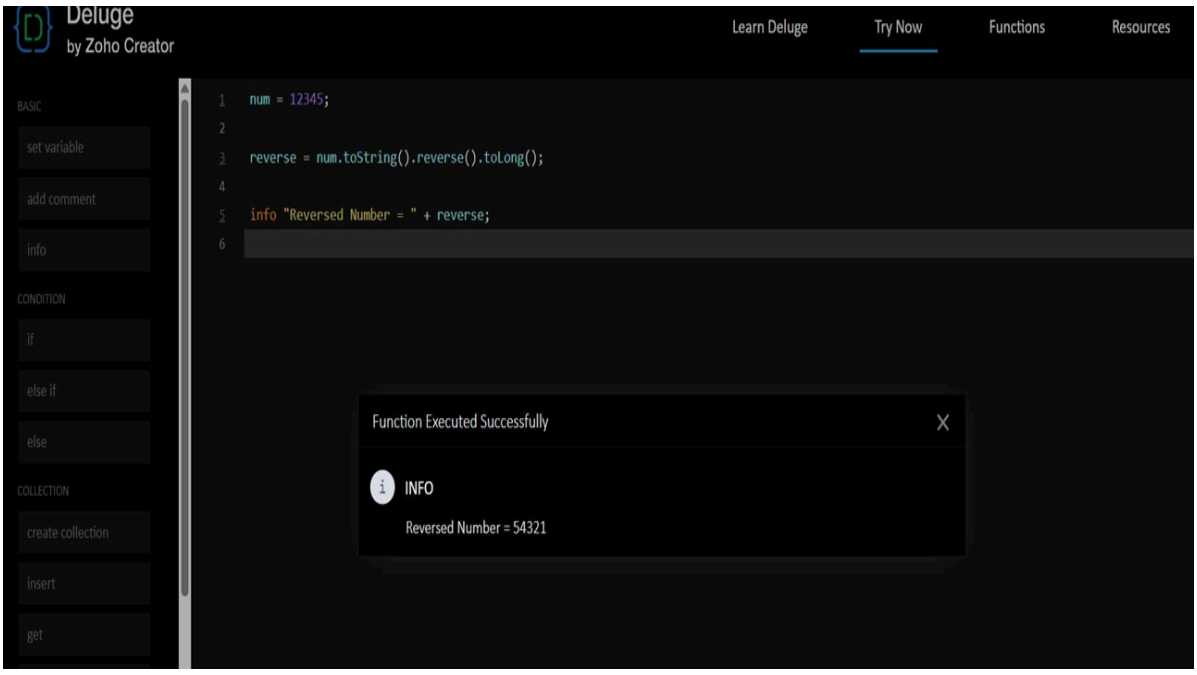


Figure 8: Deluge operation for reversing a number.

9. Palindrome Check

The input text is reversed and compared with the original value. Matching values indicate that the input is a palindrome.

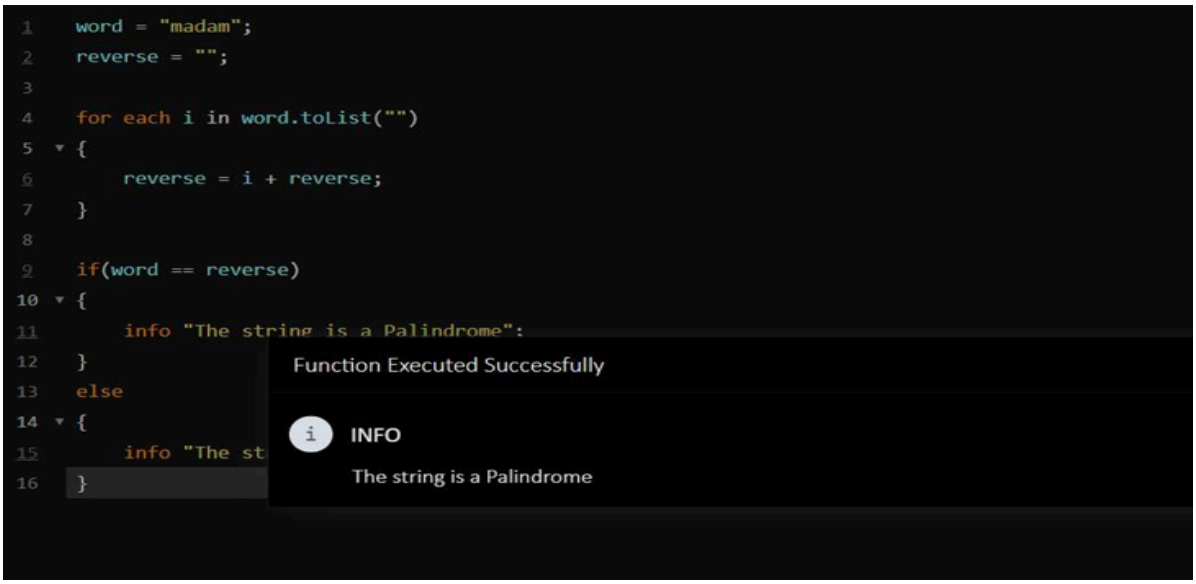


Figure 9: Deluge string operation used for palindrome verification.

10. Grade Calculation

The entered mark is evaluated against a sequence of conditions to assign the corresponding grade.

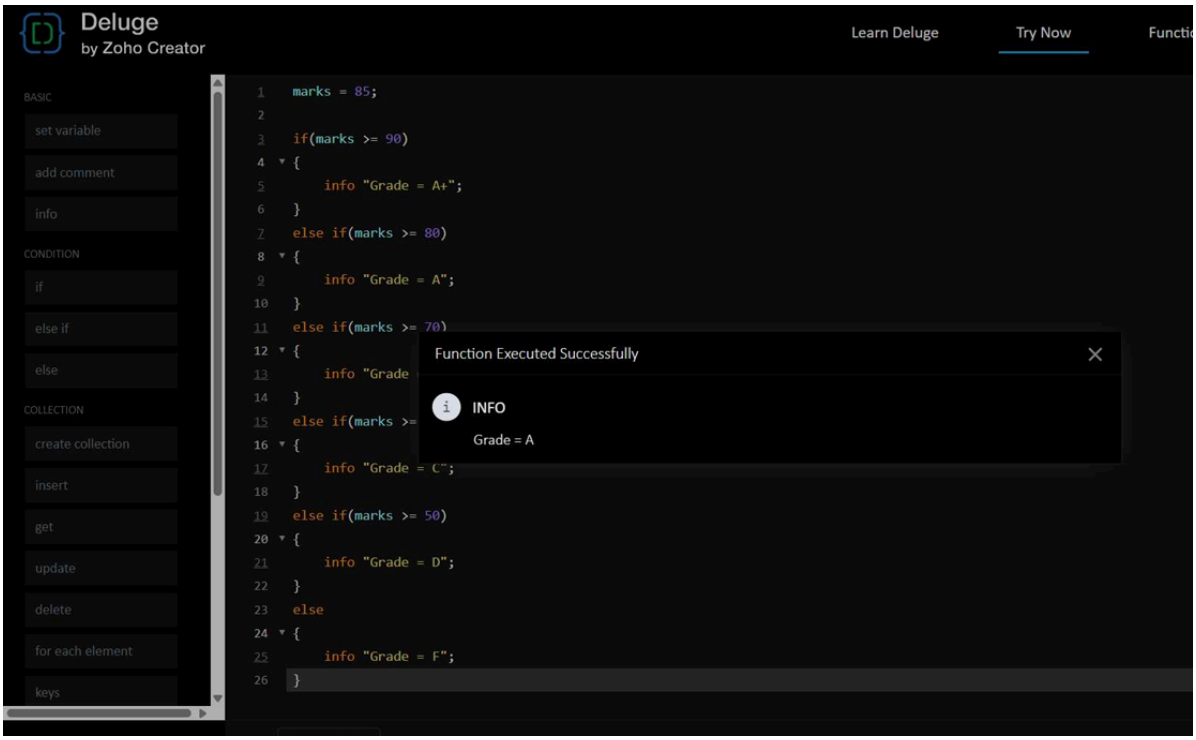


Figure 10: Deluge conditional statements used for grade determination.

6. Result

Ten Deluge functions were successfully implemented and executed in Zoho Creator. The programs demonstrated arithmetic processing, conditional branching, iteration, number manipulation, string comparison, and basic validation using Deluge scripting.